Live NBA Win Probability Predictor

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DSCI 510

Introduction

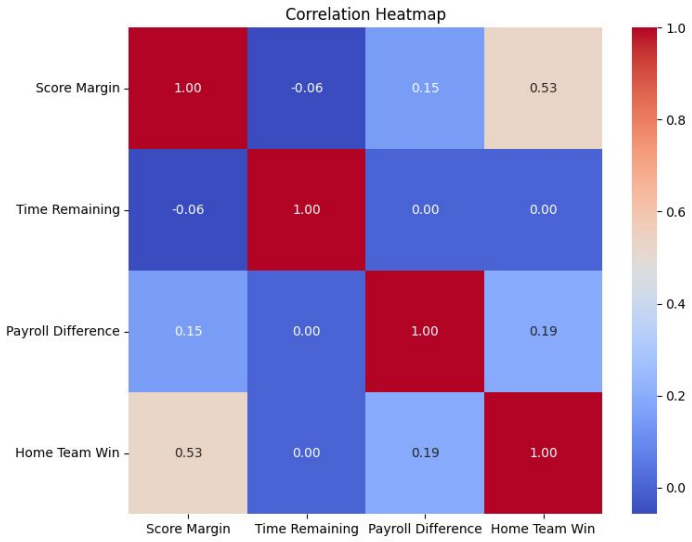
The goal of this project is to provide accurate win probabilities for NBA games as they happen in real time, while also finding insights into what determines likelihood of a team winning.

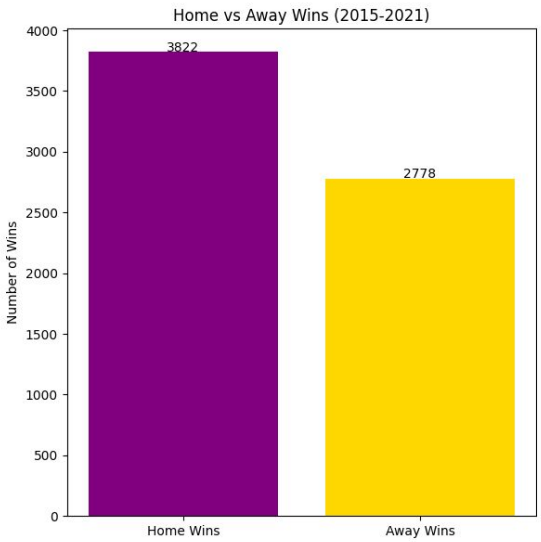
- It utilizes a machine learning model that predicts the probabilities of the home team winning given a chosen game
- It processes each play in an NBA game via a logistic regression model
- For each play, the model predicts the outcome of the game
- Three main parameters are used in the model to try to best calculate win probability during the game:
    - Score margin, team salaries, and time remaining
- Ultimately, the final aim is to compare this model's performance to other probability models

Data Sources

Datasource	Description	Type	Purpose
Historical NBA Play-by-Play Data (2015-2021), Kaggle	More than three million NBA plays from the 2015-2021 seasons	CSV	Historical data to train the model
Unofficial ESPN API	Live NBA play-by-play data	LIVE API	Used to apply the model, update real-time probabilities, and compare to our live models
NBA Team Stats, Kaggle	NBA team win percentage from 2000-2023	CSV	To find impact of parameters on winning over time
NBA Players & Team Data, Kaggle	NBA team salary information from 1990-2023	CSV	To observe effect of payroll on game outcome

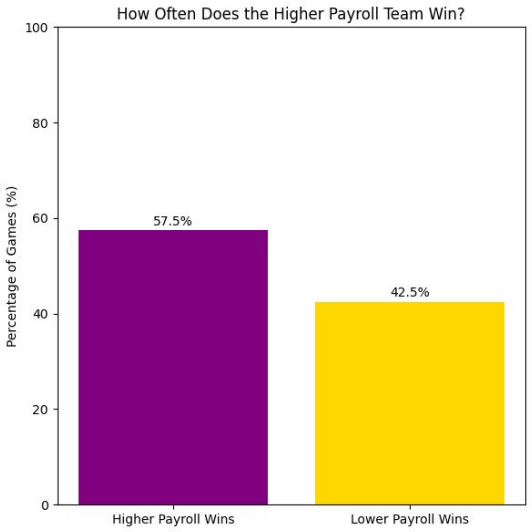
Results

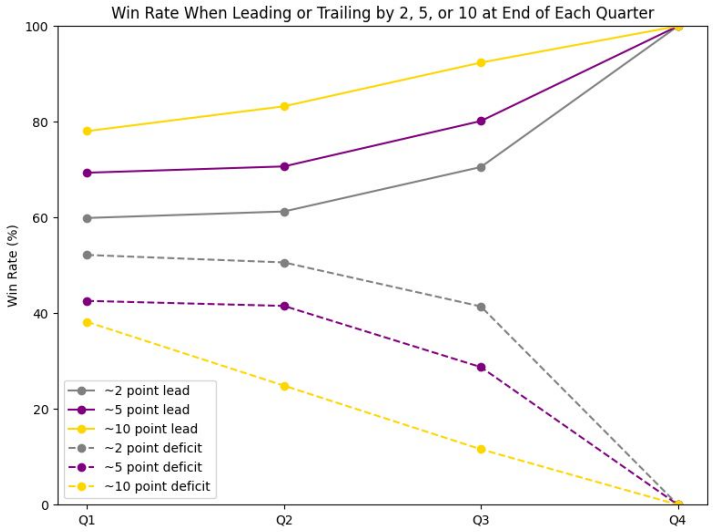




- Correlation heatmap reveals a slightly strong positive relationship between scoring margin and the home team winning
- There is only a very weak positive relationship between having a higher payroll and the home team prevailing
- From NBA seasons 2015-2021, independent of other variables, the home team won roughly 58% of the games

Results





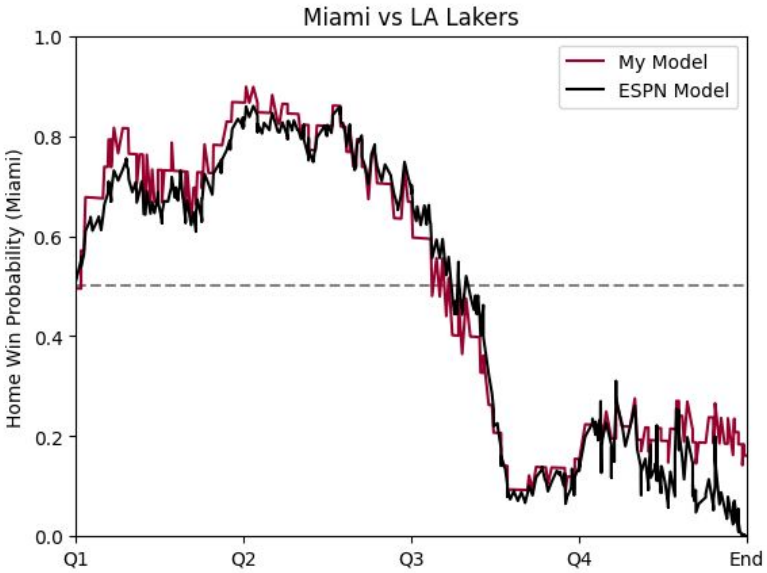
- There is a substantial difference on win rate when looking at payrolls, as the team with the higher payroll in the game wins 57.5% of the time
- The fourth quarter is when it matters: win probabilities change rapidly in the fourth quarter of games
- The impact of home court is prominent as a home team trailing by two at half time still has 50.44% chance to win, according to the model

Results

Pick a Date mm/dd/yyyy

Choose game:

- Orlando Magic at Charlotte Hornets
- Detroit Pistons at Washington Wizards
- Los Angeles Lakers at Miami Heat
- Cleveland Cavaliers at Chicago Bulls
- LA Clippers at New Orleans Pelicans
- Phoenix Suns at San Antonio Spurs
- Milwaukee Bucks at Utah Jazz
- Philadelphia 76ers at Sacramento Kings



- The two models differ by an average of 5.08%, but follow a similar pattern
- Each model tracks the game flows, big plays, and team runs
- Biggest play stat: tied game, 9:31 in Q3, Luka Doncic 25-foot three pointer
- ESPN's model is more complex, while my linear regression model uses three main variables

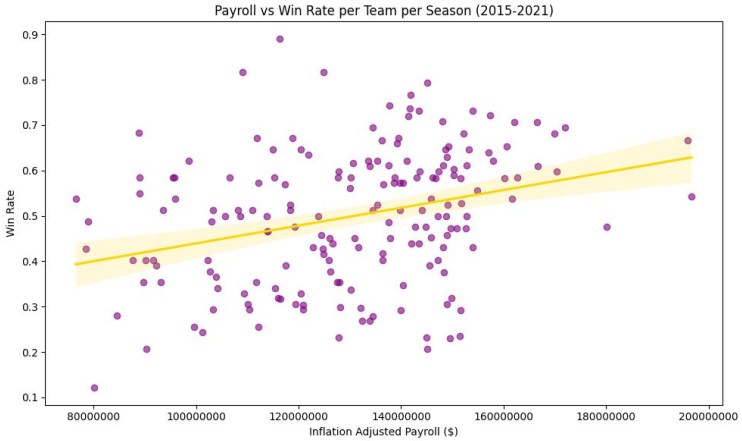
Challenges

Difficulties:

- Overall difficulty finding datasets with corresponding years
- Unable to access scheduling data and active team records
- First time trying to implement machine learning model and Matplotlib
- Mapping across multiple datasets, years, and team names
- Challenge in understanding the payroll data, heavily impactful teams

Future Scope:

- Model never fully ends at 100%
- Would like to implement more advanced stats, such as NetRating
- Observe the impact of Covid-19 and the ‘bubble’ season



THANK YOU!